Paul MacLean

www.paul-maclean.com Pittsburgh, PA
macleanp98@gmail.com 412-389-7756

Experience

Agile 5 Technologies August 2024 – November 2025
Modernization: Full Stack Developer

- Built NestJS endpoints and SvelteKit components for CRUD, features, and OIDC authentication
- Established a modernized SSR frontend for regulations.gov using SvelteKit and TailwindCSS
- Ensured product was tested (Vitest + Playwright), branded, accessible (508), and secure

Agile 5 Technologies September 2023 – August 2024
Legacy: Frontend Developer

- Spearheaded new comment clustering featureset, saving thousands of hours of manual review
- Remediated direct and transitive package vulnerabilities via upgrades and shims
- Maintained legacy Ember and JS Spring Boot microservices for regulations.gov

Agile 5 Technologies November 2022 – September 2023
Innovation Lab: Full Stack Developer

- Designed and implemented a rule engine that assesses NIST 800-53 cloud compliance across via a NextJS/Pocketbase monolith
- Added SonarQube security scanning to CI/CD pipeline; uploaded findings to CVE warehouse

West Virginia University August 2020 – May 2022
Graduate Research Assistant

- Trained thousands of models to predict dev effort using Jira and GitHub data for NASA IV&V
- Engineered bespoke Python pipeline to extract and classify features from open-source repos
- Graded Analysis of Algorithms fundamentals (data structures, complexity, graph algorithms)

U.S. Steel May 2019 – September 2019
Process Control Engineer (Internship)

- Trusted to debug long-lived Fortran programs running in VMS, on on-prem Itanium hardware
- Learned the workings of the steelmaking process, and how software integrates into each step

Education

MS in Computer Science (3.75) 2020 – 2022
West Virginia University, Morgantown, WV

BS in Computer Science (3.8) 2016 – 2020
California University of Pennsylvania, California, PA

Skills

Tools and Concepts

NodeJS, NestJS, NextJS, SvelteKit, Vitest, Jest, Tailwind, Spring, Tanstack, Auth Schemes, SQL, ORMs, AWS, CloudFlare, Git, BitBucket, Playwright, EmberJS, Machine Learning, Websockets, CI/CD, Microservices, Producer/Consumer

Languages

TypeScript, Python, Java, C, C++, Fortran, Lua

Consulting & Collaboration

End-to-End Mindset, Live Demos, Client Engagement, Atlassian Suite, Documentation, Troubleshooting, Deployments, Agile/Scrum

Projects

Serverless Realtime Pixel-Placing Game

Made an end-to-end realtime, collaborative pixel-placing minigame. Built with serverless CloudFlare workers, Durable Object cloud primitives, websocket streaming, and turnstile pseudo-authentication.

Personal Website

Showcases my work principles and projects: **www.paul-maclean.com**

Software Development Club Founder + President (California University of PA)

Established a community for extracurricular software development interests, focused on industry career growth. Still in operation.

Othello Boardgame Digitizer

Applied image processing techniques to transform off-angle board images into 2D arrays for virtual play. Trained neural network to classify Othello tiles as empty, white, or black.

Vulcan Transit Tracker

Built an IoT prototype for an Arduino bus tracking system for alma mater shuttle service. Interfaced with 4G module via AT commands to send coordinates to Spring backend.

Publications

Paul MacLean, Tom Devine, Dale Dzielski, Joshua Hernandez (2023). Efficacy of Reported Issue Times as a Means for Effort Estimation. CSCE 2023, IEEE. ISBN: 979-8-3503-2759-5